Engineering Thermodynamics - Laws, Cycles & Entropy

Department: Mechanical Engineering | Semester: 3 | Code: ME301
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1. Zeroth and First Laws of Thermodynamics

The First Law states that energy cannot be created or destroyed, only transformed: dQ = dU + dW. For a closed system undergoing a process, the net heat transfer equals the change in internal energy plus work done.

2. Second Law & Carnot Efficiency

Kelvin-Planck and Clausius statements establish the directional nature of heat engines and refrigerators. The maximum theoretical efficiency is defined by the Carnot engine: eta = 1 - (T_cold / T_hot).

3. Practical Power Cycles

Otto cycle (petrol engines), Diesel cycle (compression-ignition engines), and Rankine cycle (steam power plants) form the backbone of modern power generation.

Cycle Name	Working Fluid	Key Characteristics
Carnot Cycle	Ideal Gas	Two reversible isothermals and two adiabatics; maximum efficiency
Otto Cycle	Air / Gas Mixture	Constant volume heat addition; four-stroke IC engines
Rankine Cycle	Water / Steam	Phase change during boiling and condensation; thermal power generatio